



WOLLO University

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Exit multiple choice questions for

Data structure and algorithm

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Q 1 - In order traversal of binary search tree will produce –

A - unsorted list

B - reverse of input

C - sorted list

D - none of the above

Hide Answer

Q 2 - Postfix expression is just a reverse of prefix expression.

A - True

B - False

Answer : B

Explanation

Expression notations are not reverse (or so) of each other, rather operators used in the expression have different arrangements.

Hide Answer

Q 3 - The following formula will produce

$$F_n = F_{n-1} + F_{n-2}$$

A - Armstrong Number

B - Fibonacci Series

C - Euler Number

D - Prime Number

Answer : B

Explanation

Fibonacci Series generates subsequent number by adding two previous numbers.

Hide Answer

Q 4 - Minimum number of spanning tree in a connected graph is

A - n

B - n^{n-1}

C - 1

D - 0

Answer : C

Explanation

Every connected graph at least has one spanning tree.

[Hide Answer](#)

Q 5 - Which method can find if two vertices x & y have path between them?

A - Depth First Search

B - Breadth First Search

C - Both A & B

D - None A or B

Answer : C

Explanation

By using both BFS and DFS, a path between two vertices of a connected graph can be determined.

[Hide Answer](#)

Q 6 - Which of the below mentioned sorting algorithms are not stable?

A - Selection Sort

B - Bubble Sort

C - Merge Sort

D - Insertion Sort

Answer : A

Explanation

Except selection sort, all other sorting algorithms are stable.

[Hide Answer](#)

Q 7 - The Θ notation in asymptotic evaluation represents –

A - Base case

B - Average case

C - Worst case

D - NULL case

Answer : A

Explanation

Θ represents average case. O represents worst case and Ω represents base case.

Show Answer

Q 8 - All possible spanning trees of graph G

A - have same number of edges and vertices.

B - have same number of edges and but not vertices.

C - have same number of vertices but not edges.

D - depends upon algorithm being used.

Answer : A

Explanation

All possible spanning trees of graph G, have same number of edges and vertices.

Show Answer

Q 9 - A pivot element to partition unsorted list is used in

A - Merge Sort

B - Quick Sort

C - Insertion Sort

D - Selection Sort

Answer : B

Explanation

The quick sort partitions an array using pivot element and then calls itself recursively twice to sort the resulting two subarray.

Hide Answer

Q 10 - An adaptive sorting algorithm –

A - adapts to new computers.

B - takes advantage of already sorted elements.

C - takes input which is already sorted.

D - none of the above.

Answer : B

Explanation

A sorting algorithm is said to be adaptive, if it takes advantage of already 'sorted' elements in the list that is to be sorted.

Q1: What is the time complexity of insertion sort in best case?

- (A) $O(n)$
- (B) $O(n^2)$
- (C) $O(n \log_2 n)$
- (D) $O(\log_2 n)$

Q2: The time factor when determining the efficiency of algorithm is measured by

- (A) Counting microseconds
- (B) Counting the number of key operations
- (C) Counting the number of statements
- (D) Counting the kilobytes of algorithm

Q3: Which of the following algorithm design technique is used in the quick sort algorithm?

- (A) Dynamic programming
- (B) Backtracking
- (C) Divide and conquer
- (D) Greedy method

Q4: For the Quick sort algorithm, what is the time complexity of the best/worst case?

- (A) best case: $O(\log(n))$ worst case: $O(n^2)$
- (B) best case: $O(n^2)$, worst case: $O(n \log(n))$
- (C) best case: $O(n \log(n))$ worst case: $O(n^2)$
- (D) best case: $O(n^2)$, worst case: $O(n^2 \log(n))$

Q5: Binary Search Method has Worst Case Time Complexity of

- (A) $2n$
- (B) $n \log(n)$
- (C) $\log(n)$
- (D) $n * n$

Q6: The average case complexity of Insertion Sort is

- (A) $O(2^n)$

- (B) $O(n^2)$
- (C) $O(n^3)$
- (D) $O(2n)$

| Q7: An hash table with chaining as a collision resolution technique degenerates to a

- (A) Stack
- (B) Queue
- (C) Linked List
- (D) Tree

| Q8: Assume the input array is nearly sorted. Then performance of Quick sort is

- (A) Better than Average case
- (B) Worst than Average case
- (C) Same as in Average case
- (D) None of these

| Q9: A technique for direct search is

- (A) Binary Search
- (B) Linear Search
- (C) Tree Search
- (D) Hashing

| Q10: The running time $T(n)$ is given as

$$T(n) = c + T(n-1), \text{ if } n > 1$$

$$= d, \text{ if } n \leq 1$$

The order of the algorithm is

- (A) n^2
- (B) n
- (C) n^3
- (D) n^n

Answer:

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	A	B	C	C	C	B	C	B	D	B

Q11: Which is not the property of Quick sort

- (A) Inplace
- (B) Adaptive
- (C) Stable
- (D) Online

Q12: The Worst case occur in linear search algorithm when

- (A) Item is somewhere in the middle of the array
- (B) Item is not in the array at all
- (C) Item is the first element in the array
- (D) Item is the last element in the array or is not there at all

Q13: The correct matching for the following pairs is

- | | |
|--------------------------|-------------------------|
| (A) 0/1 Knapsack | (1) Greedy |
| (B) Quick sort | (2) Depth-first search |
| (C) Minimum weight | (3) Dynamic programming |
| (D) Connected Components | (4) Divide and conquer |

- (A) A-2 , B-4 , C-1, D-3
- (B) A-3 , B-4 , C-1 , D-2
- (C) A-3 , B-4 , C-2 , D-1
- (D) A-4 , B-1 , C-2 , D-3

Q14: A Bi-connected Graph Certainly does not contain any

- (A) Cycle
- (B) Path
- (C) Parallel Edges
- (D) Cut Vertex

Q15: The number of ways to multiply 4 Matrices in a Chained Matrix Multiplication

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q16: The spanning tree of connected graph with 10 vertices contains

- (A) 9 edges
- (B) 10 edges
- (C) 11 edges
- (D) 11 vertices

Q17: Consider the following statements.

- I. An algorithm is a no. of steps to be performed to solve a problem.
- II. To a given problem there may be more than one algorithm

- (A) Only I is correct
- (B) Only II is correct
- (C) Both I and II are false
- (D) Both I and II are correct

Q18: At least how many comparisons are required for merging two sorted lists of n elements each?

- (A) $2n - 1$
- (B) $n - 1$
- (C) $2n + 1$
- (D) n

Q19: For an undirected graph with n vertices and e edges, the sum of the degree of each vertex is equal to

- (A) $2n$
- (B) $(2n-1)/2$
- (C) $2e$
- (D) $e^2/2$

Q20: Which of the following shows the correct relationship?

- (A) $O(n \log(n)) < O(n)$
- (B) $O(2^n) < O(n^2)$
- (C) $O(n^3) < O(n^2 \log(n))$
- (D) $O(\log(n)) < O(n)$

Question	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Answer	D	D	B	D	D	A	D	A	C	D

Q21: Best case time complexity of heap sort technique is

- (A) Stable

- (B) $O(n^2)$
- (C) $O(n \log_2 n)$
- (D) $O(\log_2 n)$

| Q22: *The Average case occur in linear search algorithm*

- (A) When Item is somewhere in the middle of the array
- (B) When Item is not in the array at all
- (C) When Item is the last element in the array
- (D) When Item is the last element in the array or is not there at all

| Q23: *In the worst case, the number of comparisons needed to search a single linked list of length n for a given element is*

- (A) $\log(n)$
- (B) $n/2$
- (C) $\log(n/2)-1$
- (D) n

| Q24: *N-Queens Problem can be solved easily by*

- (A) Dynamic Programming
- (B) Backtracking Method
- (C) Greedy Method
- (D) Divide and Conquer Method.

| Q25: *Worst case Time Complexity of Linear Search is*

- (A) $O(n)$
- (B) $O(\log(n))$
- (C) $O(1)$
- (D) $O(n \log(n))$

| Q26: *Worst case Time Complexity of Heap Sort Algorithm is*

- (A) $O(\log(n))^2$
- (B) $O(n)$
- (C) $O(n^2)$
- (D) $O(n \log(n))$

| Q27: *Analyzing of an algorithm involves*

- (A) Evaluating the complexity only
- (B) Validating the algorithm Only
- (C) Both Validating the algorithm and Evaluating the Complexity
- (D) None of these

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Q28: Which of the following data structure may be used to aid implementation of radix sort?

- (A) Stack
- (B) Heap
- (C) Binary Search Tree
- (D) Queue

Q29: Which of the following sorting methods would be most suitable for sorting a list which is almost sorted

- (A) Bubble Sort
- (B) Selection Sort
- (C) Quick Sort
- (D) Merge Sort

Q30: Breadth First Traversal (BFS) is a method to traverse

- (A) Graph using shortest path
- (B) All successors of a visited node before any successors of any of those successors
- (C) A single path of the graph as far as it can go
- (D) None of these

Answer:

Question	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
Answer	C	A	D	B	A	D	C	D	A	B

Q31: The correct matching for the following pairs is

- (A) 0/1 Knapsack (1) Greedy
- (B) Multiply numbers (2) Depth-first search
- (C) Minimum weight (3) Dynamic programming
- (D) Connected Components (4) Divide and conquer

- (A) A-2 , B-4 , C-1, D-3
- (B) A-3 , B-4 , C-1 , D-2
- (C) A-3 , B-4 , C-2 , D-1
- (D) A-4 , B-1 , C-2 , D-3

Q32: The complexity of Bubble sort algorithm is

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(n \log n)$

Q33: For the bubble sort algorithm, what is the time complexity of the best/worst case? (assume that the computation stops as soon as no more swaps in one pass)

- (A) best case: $O(n)$ worst case: $O(n*n)$
- (B) best case: $O(n)$ worst case: $O(n*\log(n))$
- (C) best case: $O(1)$ worst case: $O(n)$
- (D) best case: $O(n*\log(n))$ worst case: $O(n*n)$

Q34: Max Heap Tree has following Property.

- (A) Right Child $>$ Parent
- (B) Both Children $>$ Parent
- (C) Both Children $<$ Parent
- (D) None of above

Q35: Breadth First Search method uses

- (A) Stack
- (B) Queue
- (C) Hash Table
- (D) None of these

Q36: In a heap tree bottom level should be filled

- (A) From left to right.
- (B) From right to left.
- (C) Completely.
- (D) None of these

Q37: The running time of an algorithm means

- (A) No. of Primitive Operations(Steps) executed in a machine-independent manner
- (B) Time taken on a Standard Computer to execute the program
- (C) Time taken by the algorithm on a particular input size.
- (D) None of these

Q38: What is the preferred form of representation of Dense graph?

- (A) Adjacency Matrix
- (B) Adjacency List

- (C) Incidence Matrix
- (D) None of these

Q39: A BST is traversed in the following order recursively: Right, root, left. The output sequence will be in

- (A) Ascending order
- (B) Descending order
- (C) Bitomic sequence
- (D) No specific order

Q40: The way a card game player arranges his cards as he picks them up one by one, is an example of

- (A) Bubble Sort
- (B) Selection Sort
- (C) Insertion Sort
- (D) Merge Sort

Answer:

Question	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
Answer	B	C	A	C	B	A	A	A	B	C

Q41: Which of the following sorting algorithm has minimum worst case time complexity?

- (A) Selection sort
- (B) Bubble sort
- (C) Quick sort
- (D) Merge sort

Q42: An algorithm is made up of 2 modules M1 and M2. If order of M1 is $f(n)$ and M2 is $g(n)$ then the order of the algorithm is

- (A) $\max(f(n), g(n))$
- (B) $\min(f(n), g(n))$
- (C) $f(n) + g(n)$
- (D) $f(n) \times g(n)$

Q43: When we say the order of a tree is M, we mean

- (A) Every non-leaf node must have M subtrees
- (B) Every non-leaf node must have M keys
- (C) Every non-leaf node can have at most M subtrees
- (D) The Height of the tree is M.

Q44: *Heap Tree is*

- (A) Complete Binary Tree
- (B) Cyclic Graph
- (C) 3-way Search Tree
- (D) Forest

Q45: *Minimum no. of states to Search/Recognize string “nano” with Finite Automata is*

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q46: *The height of heap tree of N data is*

- (A) $O(N)$
- (B) $O(N \log(N))$
- (C) $O(\log(N))$
- (D) $O(2^N)$

Q47: *$f(n) = \theta(g(n))$ is*

- (A) $g(n)$ is asymptotic upper bound for $f(n)$
- (B) $g(n)$ is asymptotic tight bound for $f(n)$
- (C) $g(n)$ is asymptotic lower bound for $f(n)$
- (D) None of these

Q48: *What is the running time of an efficient algorithm to find an Euler Tour in a graph if one exists?*

- (A) $O(|V|)$
- (B) $O(|E|)$
- (C) $O(|V| |E|)$
- (D) $O(|V|^2)$

Q49: *Quick sort is also known as*

- (A) merge sort
- (B) heap sort
- (C) bubble sort
- (D) None of these

Q50: *The concept of Order (Big O) is important because*

- (A) It can be used to decide the best algorithm that solves a given problem
- (B) It determine the Maximum space occupied by the algorithm

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- (C) It is the lower bound of the growth rate of the algorithm
- (D) None of these

Answer:

Question	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
Answer	D	A	C	A	D	C	B	B	D	A

Q51: Longest Common Subsequence Problem can be optimally solved by

- (A) Greedy Method
- (B) Divide & Conquer
- (C) Dynamic Programming
- (D) None of these

Q52: There are 4 different algorithms A1, A2, A3, A4 to solve a given problem with the order $\log(n)$, $\log(\log(n))$, $n\log(n)$, $n / \log(n)$ respectively. Which is the best algorithm?

- (A) A1
- (B) A2
- (C) A3
- (D) A4

Q53: Worst case Time Complexity of Merge Sort is

- (A) $O(n^2)$
- (B) $O(n \log(n))$
- (C) $O(2 \log(n))$
- (D) $O(n^2 \log(n))$

Q54: Dynamic Programming Method is not suitable to solve

- (A) 0/1 Knapsack Problem
- (B) Making Change Problem
- (C) Binomial Co-efficient
- (D) Fractional Knapsack Problem

Q55: Following is a NP Problem related to Graph

- (A) To find Longest Path
- (B) To find Shortest Path
- (C) To find Minimum Spanning Tree
- (D) To find a Cycle

Q56: A sorted array in ascending order is

- (A) MAX Heap

- (B) MIN Heap
- (C) Not a Heap Tree
- (D) None of these

Q57: $f(n) = \Omega(g(n))$ is

- (A) $g(n)$ is asymptotic upper bound for $f(n)$
- (B) $g(n)$ is asymptotic tight bound for $f(n)$
- (C) $g(n)$ is asymptotic lower bound for $f(n)$
- (D) None of these

Q58: Following Problem(s) is NP Complete

- (A) Vertex Cover
- (B) SAT(Satisfiable)
- (C) K-Colouring
- (D) All of these

Q59: An algorithm is made up of two independent time complexities $f(n)$ and $g(n)$. Then the complexities of the algorithm is in the order of

- (A) $f(n) \times g(n)$
- (B) $\max(f(n), g(n))$
- (C) $\min(f(n), g(n))$
- (D) $f(n) + g(n)$

Q60: The common implementation of a heap is

- (A) Array
- (B) Linked List
- (C) Multilinked Structure
- (D) Doubly Linked List

Answer:

Question	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60
Answer	C	B	B	D	A	B	C	D	B	A

Q61: The goal of hashing is to produce a search that takes

- (A) $O(1)$ time
- (B) $O(n^2)$ time
- (C) $O(\log n)$ time
- (D) $O(n \log n)$ time

Q62: The time complexity of an algorithm $T(n)$, where n is the input size, is given by $T(n) = T(n-1) + (1/n)$ if $n > 1$. The order of this algorithm is

- (A) $\log n$
- (B) n
- (C) n^2
- (D) n^n

Q63: The number of edges in a Minimum Spanning Tree of a connected Graph G with n vertices is

- (A) $n \log(n)$
- (B) $n!$
- (C) $n*(n-1)/2$
- (D) $n - 1$

Q64: Krushkal's Algorithm uses

- (A) Greedy Method
- (B) Divide and Conquer Method.
- (C) Dynamic Programming
- (D) Branch & Bound Method

Q65: The "Principle of Optimality" is used in

- (A) Greedy Method
- (B) Backtracking
- (C) Dynamic Programming
- (D) Branch & Bound

Q66: Which of the following algorithm do not have a time complexity of $O(n^2)$

- (A) Bubble Sort
- (B) Shell Sort
- (C) Radix Sort
- (D) Quick Sort

Q67: $f(n) = O(g(n))$ is

- (A) $g(n)$ is asymptotic upper bound for $f(n)$
- (B) $g(n)$ is asymptotic tight bound for $f(n)$
- (C) $g(n)$ is asymptotic lower bound for $f(n)$
- (D) None of these

Q68: Graphs are represented using

- (A) Adjacency tree
- (B) Adjacency linked list
- (C) Adjacency Graph

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- (D) Adjacency Queue

Q69: The goal of hashing is to produce a search that takes

- (A) $O(1)$ time
- (B) $O(n^2)$ time
- (C) $O(\log n)$ time
- (D) $O(n \log n)$ time

Q70: The depth-first search

- (A) Uses a queue
- (B) Uses a stack
- (C) Searches a tree
- (D) Searches a linked list

Answer:

Question	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70
Answer	A	A	D	A	C	C	A	B	A	B

Q71: Consider the Array: 26, 35, 1, 49, 54, 30, 99. How array will look like after 2 (two) iterations of Bubble Sort?

- (A) 26, 1, 35, 49, 54, 30, 99
- (B) 26, 1, 35, 49, 30, 54, 99
- (C) 1, 26, 35, 30, 49, 54, 99
- (D) None of these

Q72: The running time of an algorithm is given by $T(n) = T(n-1) + T(n-2) - T(n-3)$, if $n > 3$, and n , otherwise.

- (A) n
- (B) $\log n$
- (C) n^n
- (D) n^2

Q73: Time Complexity of Tower of Hanoi Problem (Recursive) with n disks is

- (A) n^2
- (B) 2^n
- (C) 3^n
- (D) n^3

Q74: NP Problem is

- (A) Polynomial Problem
- (B) Non Deterministic Polynomial Problem

- (C) Easy to Solve problem
- (D) All of Above

Q75: *Worst case Time Complexity of Multiplication of two $n \times n$ Matrix is*

- (A) n^2
- (B) $n^2 \log(n)$
- (C) $n \log(n)$
- (D) n^3

Q76: *Consider the Array: 26, 35, 11, 49, 54, 30, 80. How array will look like after 2 (two) iterations of Bubble Sort?*

- (A) 26, 11, 35, 49, 54, 30, 80
- (B) 26, 11, 35, 49, 30, 54, 80
- (C) 11, 26, 35, 30, 49, 54, 80
- (D) None of these

Q77: *$f(n) = \theta(g(n))$ implies*

- (A) $f(n) = O(g(n))$ only
- (B) $f(n) = \Omega(g(n))$ only
- (C) $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$
- (D) None of these

Q78: *Infinite recursion leads to*

- (A) Overflow of run-time stack
- (B) Underflow of registers usage
- (C) Overflow of I/O cycles
- (D) Underflow of run-time stack

Q79: *The time required to find shortest path in a graph with n vertices and e edges is*

- (A) $O(e)$
- (B) $O(n)$
- (C) $O(e^2)$
- (D) $O(n^2)$

Q80: *For NP-Complete problem*

- (A) Several Polynomial time algorithms are available.
- (B) Polynomial Time algorithms are not exist, hence can not be discovered
- (C) No Polynomial Time algorithm is discovered yet
- (D) None of Above

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Question	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80
Answer	C	A	B	B	D	B	C	A	D	C

Q81: The depth of a complete binary tree is given by

- (A) $D_n = n \log_2 n$
- (B) $D_n = n \log_2 n + 1$
- (C) $D_n = \log_2 n$
- (D) $D_n = \log_2 n + 1$

Q82: Which of the following algorithms solves the all-pair shortest path problem?

- (A) Dijkstra's algorithm
- (B) Floyd's algorithm
- (C) Prim's algorithm
- (D) Warshall's algorithm

Q83: Total Degree of a Triangle is

- (A) 6
- (B) 4
- (C) 3
- (D) 2

Q84: For a same problem If Time Complexity of Recursive algorithm is x and Time Complexity of Iterative algorithm is y, then

- (A) $x < y$
- (B) $x > y$
- (C) $x = y$
- (D) can not be said

Q85: Worst case Time Complexity of Addition of two $n \times n$ Matrix is

- (A) n^2
- (B) $n^2 \log(n)$
- (C) $n \log(n)$
- (D) n^3

Q86: In a Selection Sort of n elements, how many times is the swap function called in the complete execution of the algorithm?

- (A) 1
- (B) $n - 1$
- (C) $n \log(n)$
- (D) n^2

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Q87: $T(n) = 2T(n/2) + k.n$, where k is constant, then $T(n)$ is equal to

- (A) $O(n \log(n))$
- (B) $O(\log(n))$
- (C) $O(n)$
- (D) $O(n^2)$

Q88: The running time for creating a heap of size n is

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(n^2)$

Q89: Search tables used by compilers for efficient searching generally use

- (A) Hash Tables
- (B) Linear lists of Records
- (C) Binary Search Tables
- (D) Binary Search Trees

Q90: If $P(n) = n^2$ then what is correct?

- (A) $P(n) = n^3$
- (B) $P(n) = n^4$
- (C) $P(n) = n^2$
- (D) All of the above

Answer:

Question	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90
Answer	D	B	A	B	A	B	A	C	A	D

Q91: Which of the following sorting algorithm is of divide-and-conquer type?

- (A) Bubble sort
- (B) Insertion sort
- (C) Quick sort
- (D) All of above

Q92: For merging two sorted lists of sizes m and n into a sorted list of size $m + n$, we require comparisons of

- (A) $O(m)$
- (B) $O(n)$
- (C) $O(m + n)$
- (D) $O(\log(m) + \log(n))$

Q93: Travelling Salesman Problem is

- (A) NP Problem
- (B) P Problem
- (C) Halting Problem
- (D) All of Above

Q94: `for(i=1 ; i<=n ; i++) {printf("**"); }` The above C code has Time Complexity of

- (A) 1
- (B) $\log(n)$
- (C) $n \log(n)$
- (D) n

Q95: The running time of the following sorting algorithm depends on whether the partitioning is balanced or unbalanced

- (A) Insertion sort
- (B) Selection Sort
- (C) Quick Sort
- (D) Merge Sort

Q96: Which of the following sorts have a $O(n \log(n))$ Worst case performance?

- (A) Insertion Sort
- (B) Quick Sort
- (C) Heap Sort
- (D) None of these

Q97: Time Complexity of Recursive Fibonacci algorithm is

- (A) $O(n^2)$
- (B) $O(n^3)$
- (C) $O(n^2 \log(n))$
- (D) $O(C^n)$, where C is a constant

Q98: Recursive procedures are implemented by using ____ data structure.

- (A) Queues
- (B) Stacks
- (C) Linked Lists
- (D) Strings

Q99: For merging two sorted lists of size m and n into a sorted size of $m+n$, we require comparisons of

- (A) $O(m)$
- (B) $O(n)$

- (C) $O(m+n)$
- (D) $O(\log(m) + \log(n))$

Q100: An algorithm

- (A) Is a finite set of instructions
- (B) Any well defined computational procedure
- (C) Zero or more quantities are externally supplied and at least one quantity is produced.
- (D) All are correct

Answer:

Question	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
Answer	C	C	A	D	C	C	D	B	C	D

Q101: Which of the following is type of amortized analysis?

- (A) Aggregate Method
- (B) Accounting Method
- (C) Potential Method
- (D) All of these

Q102: Divide and Conquer method is not suitable to solve Problem like

- (A) Merge Sort
- (B) Finding Minimum Spanning Tree
- (C) Exponentiation
- (D) Binary Search

Q103: Level Ordered Traversal of a Rooted Tree can be done by starting from the root and performing

- (A) In-order traversal
- (B) Preorder traversal
- (C) Breadth First Search
- (D) Depth First Search

Q104: Which of the following is not a Graph Traversal technique?

- (A) In-order Traversal
- (B) BFS
- (C) DFS
- (D) None of These

Q105: If there is an NP-Complete Language L whose Complement is in NP, Then the Complement of any Language in NP is in

- (A) P
- (B) NP
- (C) Both A and B
- (D) None of these

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Q106: Applying Prim's algorithm to find Minimum Spanning Tree is more suitable for

- (A) Sparse Matrix
- (B) Dense Graph
- (C) Sparse Graph
- (D) None of these

Q107: Min-max Principle is used for

- (A) Gaming
- (B) Finding Minimum Spanning Tree
- (C) Finding Non Leaf Node
- (D) Divide and Conquer

Q108: Simple Merge merges

- (A) Two sorted array
- (B) Two unsorted array
- (C) Two Trees
- (D) None of these

Answer:

Question	Q101	Q102	Q103	Q104	Q105	Q106	Q107	Q108	Q109	Q110
Answer	D	B	C	A	B	B	A	A		

Q111: Divide and Conquer is a general design paradigm does not consist the following part

- (A) Divide
- (B) Recursion
- (C) Iteration
- (D) Conquer

Q112: "Chained Matrix Multiplication" can be solved by

- (A) Dynamic Programming
- (B) Greedy Method
- (C) Branch & Bound
- (D) Backtracking

Q113: Which of the following Pairs of traversals define Binary Tree uniquely?

- (A) Pre-order and Post-order
- (B) In-order and Pre-order
- (C) Level –order and Post-order
- (D) None of these

Q114: Greedy algorithms have following characteristics

- (A) Objective function
- (B) Feasible solution
- (C) Selection Function
- (D) All of these

Q115: Both P and NP are closed under the operation of

- (A) Union
- (B) Intersection
- (C) Concatenation
- (D) Kleene's

Q116: Algorithm analysis is useful for finding

- (A) Space Complexity
- (B) Time Complexity
- (C) Correctness
- (D) All of these

Q117: The approach used by linear search is

- (A) Greedy
- (B) Divide & Conquer
- (C) Probabilistic
- (D) Brute-Force

Q118: In Dynamic Programming after table is created of m rows and n columns, finding a particular solution takes order of

- (A) $\log(m)$
- (B) $m \cdot n$
- (C) $m+n$
- (D) n^2

Answer:

Question	Q111	Q112	Q113	Q114	Q115	Q116	Q117	Q118	Q119	Q120
Answer	C	A	B	D	D	D	D	C		

Q121: $f(n) = n^2$, $g(n) = n^3$, then which of the following is not correct?

- (A) $g(n) = \Omega(n)$
- (B) $g(n) = O(n^2)$
- (C) $g(n) = O(n^3)$
- (D) $g(n) = \Omega(f(n))$

Q122: "Longest Common Subsequence" Problem can be solved by

- (A) Greedy Method
- (B) Dynamic Programming
- (C) Backtracking
- (D) None of these

Q123: The Time Complexity of finding Transitive Closure of a Binary Relation on a set of n elements is known to be

- (A) $O(n^3)$

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- (B) $O(n)$
- (C) $O(n^2 \log n)$
- (D) $O(n^2)$

Q124: Divide & Conquer can't solve Problems like

- (A) Merge Sort
- (B) Exponentiation
- (C) Minimum Spanning Tree
- (D) Binary Search

Q125: Binary Search method uses as input

- (A) Unsorted Array
- (B) Linear Linked List
- (C) Sorted Array
- (D) Hash Table

Q126: Which of the following is not the required condition for binary search algorithm?

- (A) There must be mechanism to delete or insert elements in the list
- (B) there should be the direct access to the middle element in any sublist
- (C) The list must be sorted
- (D) None of these

Q127: One NP-Complete Problem can be reduced to another problem of NP in What time?

- (A) Polynomial Time
- (B) Exponential Time
- (C) Linear Time
- (D) Logarithmic Time

Q128: Feasible Solution in Greedy Method is

- (A) Satisfying constraints
- (B) Always optimal
- (C) Not a solution
- (D) None of these

Answer:

Question	Q121	Q122	Q123	Q124	Q125	Q126	Q127	Q128	Q129	Q130
Answer	B	B	A	C	C	A	A	A		

Q131: In the division method for creating hash function, which of the following hash table size is appropriate?

- (A) 2
- (B) 4
- (C) 7
- (D) 8

Q132: What is the minimum number of Stacks of size n required to implement a Queue of size n ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q133: In a binary max heap containing n elements, the smallest element can be found in

- (A) $\theta(1)$
- (B) $\theta(n)$
- (C) $\theta(\log(n))$
- (D) $\theta(n \log(n))$

Q134: Chained Matrix Multiplication problem can be solved by

- (A) Greedy Method
- (B) Dynamic Programming
- (C) Backtracking
- (D) None of these

Q135: Which of the method is used to solve Recurrences?

- (A) Substitution Method
- (B) Recursive Tree Method
- (C) Master Method
- (D) All of these

Q136: In Hamiltonian Cycle for n vertices, we

- (A) Can visit to same vertex two times
- (B) Can't visit same vertex more than one time
- (C) Can omit one vertex
- (D) None of these

Q137: Which of the following functions are often referred as “exponential growth function”?

- (A) 2^n , $\log(n)$
- (B) 2^n , $n!$

- (C) $n!$, $n\log(n)$
- (D) $n!$, $\log(n)$

Q138: Which of the following is not a property of algorithm?

- (A) Finiteness
- (B) Effectiveness
- (C) Unambiguousness
- (D) None of these

Answer:

Question	Q131	Q132	Q133	Q134	Q135	Q136	Q137	Q138	Q139	Q140
Answer	C	B	B	B	D	B	B	D		

Q141: Greedy Algorithms have following characteristic.

- (A) Objective function
- (B) Feasible solution
- (C) Selection function
- (D) All of these

Q142: Maximum number of edges in a n node graph is

- (A) $n(n-1)/2$
- (B) n^2
- (C) $n^2 \log(n)$
- (D) $n+1$

Q143: Which of the following sorting algorithm has minimum worst case time complexity?

- (A) Selection Sort
- (B) Bubble Sort
- (C) Quick Sort
- (D) Merge Sort

Q144: Longest Common Subsequence Problem can be solved by

- (A) Greedy Method
- (B) Divide & Conquer
- (C) Dynamic Programming
- (D) None of these

Q145: Applying Krushkal's algorithm to find Minimum Spanning Tree is more suitable for

- (A) Sparse Matrix
- (B) Dense Graph
- (C) Sparse Graph
- (D) None of these

Q146: "Preconditioning" term is used with

- (A) Linked List
- (B) Dequeue

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- (C) Graph
- (D) Priority Queue

Q147: *Branch & Bound Technique uses*

- (A) Lower Bound
- (B) Upper Bound
- (C) Both Lower Bound & Upper Bound
- (D) None of these

Answer:

Question	Q141	Q142	Q143	Q144	Q145	Q146	Q147	Q148	Q149	Q150
Answer	D	A	D	C	C	C	C			